

Cloud Computing Practical Exam

QUESTION 1:

```
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Sat Jun 13 05:10:39 2026 from 18.206.107.27
(ec2-user@ip-172-31-16-245 ~)$ sudo yum install -y docker
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.

=====
Package                                Architecture      Version            Repository          Size
=====
Installing:
docker                                x86_64            25.0.16-1.amzn2023.0.2    amazonlinux          46 M
Installing dependencies:
container-selinux                     noarch            4:2.245.0-1.amzn2023     amazonlinux          58 k
containerd                             x86_64            2.2.4-1.amzn2023.0.1      amazonlinux          24 M
iptables-libs                         x86_64            1.8.8-3.amzn2023.0.2     amazonlinux          401 k
iptables-nft                         x86_64            1.8.8-3.amzn2023.0.2     amazonlinux          183 k
libcgroup                             x86_64            3.0-1.amzn2023.0.1       amazonlinux           75 k
libnetfilter_conntrack               x86_64            1.0.8-2.amzn2023.0.2     amazonlinux          58 k
libnftnl                             x86_64            1.0.1-19.amzn2023.0.2    amazonlinux          30 k
libnftlink                           x86_64            1.2.2-2.amzn2023.0.2     amazonlinux          84 k
pigz                                 x86_64            2.5-1.amzn2023.0.4       amazonlinux           83 k
runc                                  x86_64            1.3.5-1.amzn2023.0.2     amazonlinux          3.9 M
=====

Transaction Summary
=====
Install 11 Packages

Total download size: 76 M
Installed size: 284 M
Downloading Packages:
(1/11): container-selinux-2.245.0-1.amzn2023.noarch.rpm                    1.6 MB/s | 58 kB  00:00

i-0438b30cbd9af20d7 (cloud-prac-instance)
PublicIPs: 54.147.13.97 PrivateIPs: 172.31.16.245

CloudShell Feedback Console Mobile App
```

```
Installing : libnftnl-1.2.2-2.amzn2023.0.2.x86_64                                4/11
Installing : libnftnl-1.2.2-2.amzn2023.0.2.x86_64                                5/11
Installing : libnftnl-1.2.2-2.amzn2023.0.2.x86_64                                6/11
Installing : iptables-libs-1.8.8-3.amzn2023.0.2.x86_64                          7/11
Installing : iptables-nft-1.8.8-3.amzn2023.0.2.x86_64                          8/11
Running scriptlet: iptables-nft-1.8.8-3.amzn2023.0.2.x86_64                    9/11
Installing : libcgroup-3.0-1.amzn2023.0.1.x86_64                               9/11
Running scriptlet: container-selinux-4:2.245.0-1.amzn2023.noarch                10/11
Installing : container-selinux-4:2.245.0-1.amzn2023.noarch                    10/11
Running scriptlet: container-selinux-4:2.245.0-1.amzn2023.noarch                10/11
Running scriptlet: docker-25.0.16-1.amzn2023.0.2.x86_64                      11/11
Installing : docker-25.0.16-1.amzn2023.0.2.x86_64                            11/11
Running scriptlet: docker-25.0.16-1.amzn2023.0.2.x86_64                      11/11
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /usr/lib/systemd/system/docker.socket.

Running scriptlet: container-selinux-4:2.245.0-1.amzn2023.noarch                11/11
Running scriptlet: docker-25.0.16-1.amzn2023.0.2.x86_64                      11/11
Verifying : container-selinux-4:2.245.0-1.amzn2023.noarch                    1/11
Verifying : containerd-2.2.4-1.amzn2023.0.1.x86_64                          2/11
Verifying : docker-25.0.16-1.amzn2023.0.2.x86_64                          3/11
Verifying : iptables-libs-1.8.8-3.amzn2023.0.2.x86_64                      4/11
Verifying : iptables-nft-1.8.8-3.amzn2023.0.2.x86_64                      5/11
Verifying : libcgroup-3.0-1.amzn2023.0.1.x86_64                            6/11
Verifying : libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64              7/11
Verifying : libnftnl-1.0.1-19.amzn2023.0.2.x86_64                        8/11
Verifying : libnftnl-1.2.2-2.amzn2023.0.2.x86_64                        9/11
Verifying : pigz-2.5-1.amzn2023.0.4.x86_64                               10/11
Verifying : runc-1.3.5-1.amzn2023.0.2.x86_64                             11/11

Installed:
container-selinux-4:2.245.0-1.amzn2023.noarch    containerd-2.2.4-1.amzn2023.0.1.x86_64    docker-25.0.16-1.amzn2023.0.2.x86_64
iptables-libs-1.8.8-3.amzn2023.0.2.x86_64      iptables-nft-1.8.8-3.amzn2023.0.2.x86_64  libcgroup-3.0-1.amzn2023.0.1.x86_64
libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64  libnftnl-1.0.1-19.amzn2023.0.2.x86_64  libnftnl-1.2.2-2.amzn2023.0.2.x86_64
pigz-2.5-1.amzn2023.0.4.x86_64                  runc-1.3.5-1.amzn2023.0.2.x86_64

Complete!
(ec2-user@ip-172-31-16-245 ~)$ docker --version
Docker version 25.0.14, build 0bab007
(ec2-user@ip-172-31-16-245 ~)$
```

```
us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-0438b30cbd9af20d7&osUser=ec2-u...
aws
Search [Option+S]
United States (N. Virginia) Naman_853

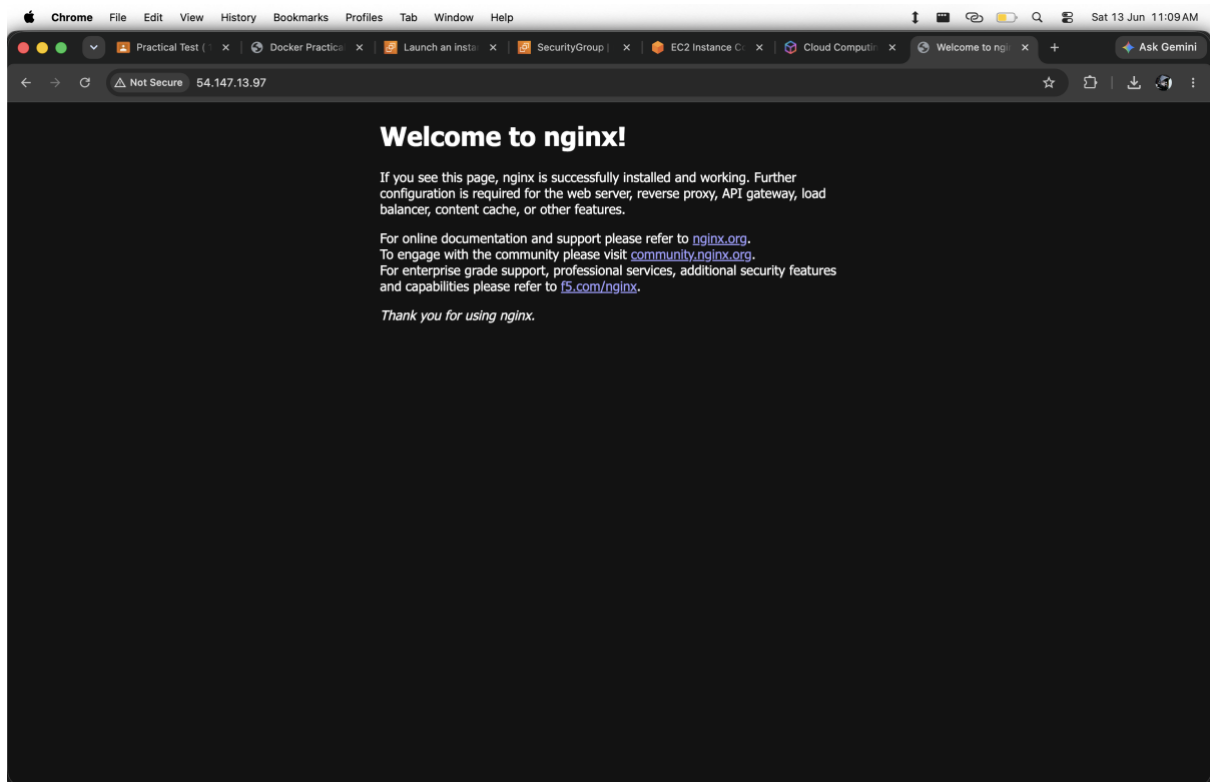
Using default tag: latest
latest: Pulling from library/nginx
Digest: sha256:608a100c71651bf5b773c99083b4a1ad7ef4b2bd05d7a7e552271e03123692ad
Status: Image is up to date for nginx:latest
docker.io/library/nginx:latest
[ec2-user@ip-172-31-16-245 ~]$ docker run nginx
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/06/13 05:33:26 [notice] 1#1: using the "epoll" event method
2026/06/13 05:33:26 [notice] 1#1: nginx/1.31.1
2026/06/13 05:33:26 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/06/13 05:33:26 [notice] 1#1: OS: Linux 6.1.174-217.345.amzn2023.x86_64
2026/06/13 05:33:26 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 32768:65536
2026/06/13 05:33:26 [notice] 1#1: start worker processes
2026/06/13 05:33:26 [notice] 1#1: start worker process 29
2026/06/13 05:33:26 [notice] 1#1: start worker process 30
^C2026/06/13 05:35:18 [notice] 1#1: signal 2 (SIGINT) received, exiting
2026/06/13 05:35:18 [notice] 29#29: exiting
2026/06/13 05:35:18 [notice] 30#30: exiting
2026/06/13 05:35:18 [notice] 30#30: exit
2026/06/13 05:35:18 [notice] 1#1: signal 14 (SIGALRM) received
2026/06/13 05:35:18 [notice] 1#1: signal 17 (SIGCHLD) received from 30
2026/06/13 05:35:18 [notice] 1#1: worker process 30 exited with code 0
2026/06/13 05:35:18 [notice] 1#1: signal 29 (SIGIO) received
2026/06/13 05:35:19 [notice] 1#1: signal 17 (SIGCHLD) received from 29
2026/06/13 05:35:19 [notice] 1#1: worker process 29 exited with code 0
2026/06/13 05:35:19 [notice] 1#1: exit
[ec2-user@ip-172-31-16-245 ~]$
[ec2-user@ip-172-31-16-245 ~]$ docker run -d --name nginx1 -p 80:80 nginx
78ba30620180a3be529662f460dd22f77d4af1ae227418041832cb33bb63cc5
[ec2-user@ip-172-31-16-245 ~]$

i-0438b30cbd9af20d7 (cloud-prac-instance)
PublicIPs: 54.147.13.97 PrivateIPs: 172.31.16.245
CloudShell Feedback Console Mobile App
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```

```
us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-0438b30cbd9af20d7&osUser=ec2-u...
aws
Search [Option+S]
United States (N. Virginia) Naman_853

[ec2-user@ip-172-31-16-245 ~]$ docker run nginx
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/06/13 05:35:41 [notice] 1#1: using the "epoll" event method
2026/06/13 05:35:41 [notice] 1#1: nginx/1.31.1
2026/06/13 05:35:41 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/06/13 05:35:41 [notice] 1#1: OS: Linux 6.1.174-217.345.amzn2023.x86_64
2026/06/13 05:35:41 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 32768:65536
2026/06/13 05:35:41 [notice] 1#1: start worker processes
2026/06/13 05:35:41 [notice] 1#1: start worker process 29
2026/06/13 05:35:41 [notice] 1#1: start worker process 30
^C2026/06/13 05:35:50 [notice] 1#1: signal 2 (SIGINT) received, exiting
2026/06/13 05:35:50 [notice] 29#29: exiting
2026/06/13 05:35:50 [notice] 30#30: exiting
2026/06/13 05:35:50 [notice] 30#30: exit
2026/06/13 05:35:50 [notice] 1#1: signal 14 (SIGALRM) received
2026/06/13 05:35:50 [notice] 1#1: signal 17 (SIGCHLD) received from 29
2026/06/13 05:35:50 [notice] 1#1: worker process 29 exited with code 0
2026/06/13 05:35:50 [notice] 1#1: signal 29 (SIGIO) received
2026/06/13 05:35:50 [notice] 1#1: signal 17 (SIGCHLD) received from 30
2026/06/13 05:35:50 [notice] 1#1: worker process 30 exited with code 0
2026/06/13 05:35:50 [notice] 1#1: exit
[ec2-user@ip-172-31-16-245 ~]$ docker run -d --name nginx1 -p 80:80 nginx
docker: Error response from daemon: Conflict. The container name "/nginx1" is already in use by container "78ba30620180a3be529662f460dd22f77d4af1ae227418041832cb33bb63cc5". You have to remove (or rename) that container to be able to reuse that name.
See 'docker run --help'.
[ec2-user@ip-172-31-16-245 ~]$ docker ps
CONTAINER ID        IMAGE               COMMAND                  CREATED        STATUS        PORTS                               NAMES
78ba30620180      nginx              "/docker-entrypoint..." 39 seconds ago Up 38 seconds 0.0.0.0:80->80/tcp, :::80->80/tcp  nginx1
01718e748032      nginx              "/docker-entrypoint..." 4 minutes ago  Up 4 minutes  80/tcp                             blissful_mclaren
[ec2-user@ip-172-31-16-245 ~]$

i-0438b30cbd9af20d7 (cloud-prac-instance)
PublicIPs: 54.147.13.97 PrivateIPs: 172.31.16.245
CloudShell Feedback Console Mobile App
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```



QUESTION 2:

```
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/06/13 05:35:41 [notice] 1#1: using the "epoll" event method
2026/06/13 05:35:41 [notice] 1#1: nginx/1.31.1
2026/06/13 05:35:41 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/06/13 05:35:41 [notice] 1#1: OS: Linux 6.1.174-217.345.amzn2023.x86_64
2026/06/13 05:35:41 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 32768:65536
2026/06/13 05:35:41 [notice] 1#1: start worker processes
2026/06/13 05:35:41 [notice] 1#1: start worker process 29
2026/06/13 05:35:41 [notice] 1#1: start worker process 30
^C2026/06/13 05:35:50 [notice] 1#1: signal 2 (SIGINT) received, exiting
2026/06/13 05:35:50 [notice] 29#29: exiting
2026/06/13 05:35:50 [notice] 29#29: exit
2026/06/13 05:35:50 [notice] 30#30: exiting
2026/06/13 05:35:50 [notice] 30#30: exit
2026/06/13 05:35:50 [notice] 1#1: signal 14 (SIGALRM) received
2026/06/13 05:35:50 [notice] 1#1: signal 17 (SIGCHLD) received from 29
2026/06/13 05:35:50 [notice] 1#1: worker process 29 exited with code 0
2026/06/13 05:35:50 [notice] 1#1: signal 29 (SIGIO) received
2026/06/13 05:35:50 [notice] 1#1: signal 17 (SIGCHLD) received from 30
2026/06/13 05:35:50 [notice] 1#1: worker process 30 exited with code 0
2026/06/13 05:35:50 [notice] 1#1: exit
[ec2-user@ip-172-31-16-245 ~]$ docker run -d --name nginx1 -p 80:80 nginx
docker: Error response from daemon: Conflict. The container name "/nginx1" is already in use by container "78ba30620180a3be529662f460dd22f77d4af1ae227418041832cb333bb63cc5". You have to remove (or rename) that container to be able to reuse that name.
See 'docker run --help'.
[ec2-user@ip-172-31-16-245 ~]$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
78ba30620180   nginx         "/docker-entrypoint..." 39 seconds ago Up 38 seconds 0.0.0.0:80->80/tcp, :::80->80/tcp   nginx1
01718e748032   nginx         "/docker-entrypoint..." 4 minutes ago  Up 4 minutes  80/tcp                             blissful_mclaren
[ec2-user@ip-172-31-16-245 ~]$ curl -s http://checkip.amazonaws.com
54.147.13.97
[ec2-user@ip-172-31-16-245 ~]$ docker volume create studentdata
studentdata
[ec2-user@ip-172-31-16-245 ~]$
```

i-0438b30cbd9af20d7 (cloud-prac-instance)

PublicIPs: 54.147.13.97 PrivateIPs: 172.31.16.245

```
2026/06/13 05:35:41 [notice] 1#1: nginx/1.31.1
2026/06/13 05:35:41 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/06/13 05:35:41 [notice] 1#1: OS: Linux 6.1.174-217.345.amzn2023.x86_64
2026/06/13 05:35:41 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 32768:65536
2026/06/13 05:35:41 [notice] 1#1: start worker processes
2026/06/13 05:35:41 [notice] 1#1: start worker process 29
2026/06/13 05:35:41 [notice] 1#1: start worker process 30
^C2026/06/13 05:35:50 [notice] 1#1: signal 2 (SIGINT) received, exiting
2026/06/13 05:35:50 [notice] 29#29: exiting
2026/06/13 05:35:50 [notice] 29#29: exit
2026/06/13 05:35:50 [notice] 30#30: exiting
2026/06/13 05:35:50 [notice] 30#30: exit
2026/06/13 05:35:50 [notice] 1#1: signal 14 (SIGALRM) received
2026/06/13 05:35:50 [notice] 1#1: signal 17 (SIGCHLD) received from 29
2026/06/13 05:35:50 [notice] 1#1: worker process 29 exited with code 0
2026/06/13 05:35:50 [notice] 1#1: signal 29 (SIGIO) received
2026/06/13 05:35:50 [notice] 1#1: signal 17 (SIGCHLD) received from 30
2026/06/13 05:35:50 [notice] 1#1: worker process 30 exited with code 0
2026/06/13 05:35:50 [notice] 1#1: exit
[ec2-user@ip-172-31-16-245 ~]$ docker run -d --name nginx1 -p 80:80 nginx
docker: Error response from daemon: Conflict. The container name "/nginx1" is already in use by container "78ba30620180a3be529662f460dd22f77d4af1ae227418041832cb333bb63cc5". You have to remove (or rename) that container to be able to reuse that name.
See 'docker run --help'.
[ec2-user@ip-172-31-16-245 ~]$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
78ba30620180   nginx         "/docker-entrypoint..." 39 seconds ago Up 38 seconds 0.0.0.0:80->80/tcp, :::80->80/tcp   nginx1
01718e748032   nginx         "/docker-entrypoint..." 4 minutes ago  Up 4 minutes  80/tcp                             blissful_mclaren
[ec2-user@ip-172-31-16-245 ~]$ curl -s http://checkip.amazonaws.com
54.147.13.97
[ec2-user@ip-172-31-16-245 ~]$ docker volume create studentdata
studentdata
[ec2-user@ip-172-31-16-245 ~]$ docker run -it --mount source=studentdata,target=/data ubuntu
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
6f5c5aa4e145: Pull complete
1e24338dd461: Pull complete
Digest: sha256:f3d28607ddd78734bb7f71f117f3c6706c666b8b76cbff7c9ff6e5718d46ff64
Status: Downloaded newer image for ubuntu:latest
root@94d6ad59c832:/#
```

i-0438b30cbd9af20d7 (cloud-prac-instance)

PublicIPs: 54.147.13.97 PrivateIPs: 172.31.16.245

```
2026/06/13 05:35:41 [notice] 1#1: nginx/1.31.1
2026/06/13 05:35:41 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/06/13 05:35:41 [notice] 1#1: OS: Linux 6.1.174-217.345.amzn2023.x86_64
2026/06/13 05:35:41 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 32768:65536
2026/06/13 05:35:41 [notice] 1#1: start worker processes
2026/06/13 05:35:41 [notice] 1#1: start worker process 29
2026/06/13 05:35:41 [notice] 1#1: start worker process 30
2026/06/13 05:35:50 [notice] 1#1: signal 2 (SIGINT) received, exiting
2026/06/13 05:35:50 [notice] 29#29: exiting
2026/06/13 05:35:50 [notice] 29#29: exit
2026/06/13 05:35:50 [notice] 30#30: exiting
2026/06/13 05:35:50 [notice] 30#30: exit
2026/06/13 05:35:50 [notice] 1#1: signal 14 (SIGALRM) received
2026/06/13 05:35:50 [notice] 1#1: signal 17 (SIGCHLD) received from 29
2026/06/13 05:35:50 [notice] 1#1: worker process 29 exited with code 0
2026/06/13 05:35:50 [notice] 1#1: signal 29 (SIGIO) received
2026/06/13 05:35:50 [notice] 1#1: signal 17 (SIGCHLD) received from 30
2026/06/13 05:35:50 [notice] 1#1: worker process 30 exited with code 0
2026/06/13 05:35:50 [notice] 1#1: exit
[ec2-user@ip-172-31-16-245 ~]$ docker run -d --name nginx1 -p 80:80 nginx
docker: Error response from daemon: Conflict. The container name "/nginx1" is already in use by container "78ba30620180a3be529662f460dd22f77d4af1ae227418041832cb333bb63cc5". You have to remove (or rename) that container to be able to reuse that name.
See 'docker run --help'.
[ec2-user@ip-172-31-16-245 ~]$ docker ps
CONTAINER ID        IMAGE               COMMAND                  CREATED              STATUS              PORTS               NAMES
78ba30620180        nginx              "/docker-entrypoint..." 39 seconds ago       Up 38 seconds       0.0.0.0:80->80/tcp, :::80->80/tcp   nginx1
01718e748032        nginx              "/docker-entrypoint..." 4 minutes ago        Up 4 minutes        80/tcp               blissful_mclaren
[ec2-user@ip-172-31-16-245 ~]$ curl -s http://checkip.amazonaws.com
54.147.13.97
[ec2-user@ip-172-31-16-245 ~]$ docker volume create studentdata
studentdata
[ec2-user@ip-172-31-16-245 ~]$ docker run -it --mount source=studentdata,target=/data ubuntu
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
6f5c5aa4e145: Pull complete
1c24335ddd46: Pull complete
Digest: sha256:f3d2860ddd78734bb7f71f117f3c6706c666b8b76cbff7c9ff6e5718d46ff64
Status: Downloaded newer image for ubuntu:latest
root@94d6ad59c832:/#
```

i-0438b30cbd9af20d7 (cloud-prac-instance)

PublicIPs: 54.147.13.97 PrivateIPs: 172.31.16.245

CloudShell Feedback Console Mobile App

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```
root@94d6ad59c832:/# echo 'Naman Thakur-24102A0008' > /data/info.txt
root@94d6ad59c832:/# cat /data/info.txt
Naman Thakur-24102A0008
root@94d6ad59c832:/#
```

i-0438b30cbd9af20d7 (cloud-prac-instance)

PublicIPs: 54.147.13.97 PrivateIPs: 172.31.16.245

CloudShell Feedback Console Mobile App

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The screenshot shows a Chrome browser window with the AWS CloudShell interface. The terminal session is running on an EC2 instance with ID i-0438b30cbd9af20d7. The user is root@94d6ad59c832. The commands executed are: `echo 'Naman Thakur-24102A0008' > /data/info.txt`, `cat /data/info.txt`, and `exit`. The output of the `cat` command is `Naman Thakur-24102A0008`. The terminal session ends with the prompt `(ec2-user@ip-172-31-16-245 ~)$`. The AWS CloudShell header shows the region as United States (N. Virginia) and the user as Naman_853. The footer shows the instance ID, public IP, and private IP.

```
root@94d6ad59c832:~# echo 'Naman Thakur-24102A0008' > /data/info.txt
root@94d6ad59c832:~# cat /data/info.txt
Naman Thakur-24102A0008
root@94d6ad59c832:~# exit
exit
(ec2-user@ip-172-31-16-245 ~)$
```

i-0438b30cbd9af20d7 (cloud-prac-instance)
PublicIPs: 54.147.13.97 PrivateIPs: 172.31.16.245

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QUESTION 3

```
aws
[ec2-user@ip-172-31-16-245 ~]$ nano Dockerfile
[ec2-user@ip-172-31-16-245 ~]$ docker build -t exam-message .
[*] Building 0.9s (5/5) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 173B
=> [internal] load metadata for docker.io/library/alpine:latest
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/1] FROM docker.io/library/alpine:latest@sha256:a2d49ea686c2adfe3c992e47dc3b5e7fa6e6b5055609400dc2acaeb241c829f4
=> => resolve docker.io/library/alpine:latest@sha256:a2d49ea686c2adfe3c992e47dc3b5e7fa6e6b5055609400dc2acaeb241c829f4
=> => extracting sha256:9b70e313681f44d32991ec943f89228bc91d7431d4a84feafc269a76e3f96a63
=> => sha256:772bee4540c41d1fec755cb8ead76db4173d38afde7b2018a08f255907bbf23 608B / 608B
=> => sha256:9b70e313681f44d32991ec943f89228bc91d7431d4a84feafc269a76e3f96a63 3.87MB / 3.87MB
=> => sha256:a2d49ea686c2adfe3c992e47dc3b5e7fa6e6b5055609400dc2acaeb241c829f4 9.22kB / 9.22kB
=> => sha256:33154315cf4402e697f065e6ec2156e292187e633908ccfed9c66279b6fa956 1.02kB / 1.02kB
=> => exporting to image
=> => exporting layers
=> => writing image sha256:61aa9f2ea4cb84c8b5e58834db39aa4b237da38265196814d9ba058720b48bd3
=> => naming to docker.io/library/exam-message
[ec2-user@ip-172-31-16-245 ~]$ docker run exam-message
Cloud Computing Practical Exam Completed Successfully
[ec2-user@ip-172-31-16-245 ~]$
```

i-0438b30cbd9af20d7 (cloud-prac-instance)
PublicIPs: 54.147.13.97 PrivateIPs: 172.31.16.245

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```
GNU nano 8.3 Dockerfile
FROM alpine
CMD echo "Cloud Computing Practical Exam Completed Successfully"
```

Read 2 lines

Help Write Out Where Is Cut Execute Location Undo Set Mark To Bracket Previous
Exit Read File Replace Paste Justify Go To Line Redo Copy Where Was Next

i-0438b30cbd9af20d7 (cloud-prac-instance)
PublicIPs: 54.147.13.97 PrivateIPs: 172.31.16.245

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Bonus:

- 1) Docker Image — a read-only template/snapshot (like a class or recipe) containing the app, dependencies, and config. It's immutable and shareable.
Docker Container — a running instance of an image (like an object or cooked meal). It has a writable layer, its own filesystem, network, and process isolation. You can start, stop, delete, and modify a container; the image stays unchanged.
- 2) Docker Volumes are used to:
 1. Persist data beyond the container's lifecycle — data survives container deletion/recreation.
 2. Share data between multiple containers (e.g., a database container and a backup container).
 3. Decouple storage from the container — the volume is managed by Docker, stored on the host filesystem (/var/lib/docker/volumes/).
 4. Backup and migrate data independently of containers.